

Experiment - 1

Data Wrangling I

Perform the following operations using Python on any open source dataset (e.g., data.csv)

1. Import all the required Python Libraries.
2. Locate an open source data from the web (e.g. <https://www.kaggle.com>). Provide a clear description of the data and its source (i.e., URL of the web site).
3. Load the Dataset into pandas data frame.
4. Data Preprocessing: check for missing values in the data using pandas `isnull()`, `describe()` function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame.
5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.
6. Turn categorical variables into quantitative variables in Python. In addition to the codes and outputs, explain every operation that you do in the above steps and explain everything that you do to import/read/scrape the data set.

```
In [1]: pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\hp\anaconda3\lib\site-packages (0.25.1)
Requirement already satisfied: pytz>=2017.2 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (2019.3)
Requirement already satisfied: python-dateutil>=2.6.1 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (2.8.0)
Requirement already satisfied: numpy>=1.13.3 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (1.16.5)
Requirement already satisfied: six>=1.5 in c:\users\hp\anaconda3\lib\site-packages (from python-dateutil>=2.6.1->pandas) (1.12.0)
Note: you may need to restart the kernel to use updated packages.
```

```
In [2]: import pandas as pd
```

```
In [4]: df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\1\StudentsPerformance.csv")
```

```
In [5]: print(df)
```

```

      gender race/ethnicity parental level of education      lunch \
0    female      group B      bachelor's degree      standard
1    female      group C      some college      standard
2    female      group B      master's degree      standard
3     male      group A      associate's degree  free/reduced
4     male      group C      some college      standard
..     ...      ...      ...      ...
995  female      group E      master's degree      standard
996   male      group C      high school  free/reduced
997  female      group C      high school  free/reduced
998  female      group D      some college      standard
999  female      group D      some college  free/reduced

      test preparation course  math score  reading score  writing score
0          none              72           72           NaN
1      completed              69           90          88.0
2          none              90           95          93.0
3          none              47           57          44.0
4          none              76           78          75.0
..     ...      ...      ...      ...
995      completed              88           99          95.0
996          none              62           55          55.0
997      completed              59           71          65.0
998      completed              68           78          77.0
999          none              77           86          86.0

```

[1000 rows x 8 columns]

```
In [7]: df.head(15)
```

Out[7]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	NaN
1	female	group C	some college	standard	completed	69	90	88.0
2	female	group B	master's degree	standard	none	90	95	93.0
3	male	group A	associate's degree	free/reduced	none	47	57	44.0
4	male	group C	some college	standard	none	76	78	75.0
5	female	group B	associate's degree	standard	none	71	83	78.0
6	female	group B	some college	standard	completed	88	95	92.0
7	male	group B	some college	free/reduced	none	40	43	39.0
8	male	group D	high school	free/reduced	completed	64	64	67.0
9	female	group B	high school	free/reduced	none	38	60	50.0
10	male	group C	associate's degree	standard	none	58	54	52.0
11	male	group D	associate's degree	standard	none	40	52	43.0
12	female	group B	high school	standard	none	65	81	73.0
13	male	group A	some college	standard	completed	78	72	70.0
14	female	group A	master's degree	standard	none	50	53	58.0

```
In [8]: df.isnull().sum()
```

```
Out[8]: gender                0
race/ethnicity              0
parental level of education  0
lunch                      0
test preparation course     0
math score                 0
reading score              0
writing score              1
dtype: int64
```

```
In [9]: print(df.describe())
```

	math score	reading score	writing score
count	1000.00000	1000.000000	999.000000
mean	66.08900	69.169000	68.048048
std	15.16308	14.600192	15.202102
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.500000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

```
In [11]: df.dtypes
```

```
Out[11]: gender                object
race/ethnicity              object
parental level of education  object
lunch                      object
test preparation course     object
math score                 int64
reading score              int64
writing score              float64
dtype: object
```

```
In [12]: df.dropna(axis=1)
```

```
Out[12]:
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score
0	female	group B	bachelor's degree	standard	none	72	72
1	female	group C	some college	standard	completed	69	90
2	female	group B	master's degree	standard	none	90	95
3	male	group A	associate's degree	free/reduced	none	47	57
4	male	group C	some college	standard	none	76	78
...	...	...	...	...	...	...	...
995	female	group E	master's degree	standard	completed	88	99
996	male	group C	high school	free/reduced	none	62	55
997	female	group C	high school	free/reduced	completed	59	71
998	female	group D	some college	standard	completed	68	78
999	female	group D	some college	free/reduced	none	77	86

1000 rows × 7 columns

```
In [13]: y = df.iloc[:, 0:1]
print(y)
```

```
gender
0    female
1    female
2    female
3     male
4     male
..     ...
995  female
996    male
997  female
998  female
999  female
```

```
[1000 rows x 1 columns]
```

```
In [18]: from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
y = le.fit_transform(y)
print(y)
```

```
[0 0 0 1 1 0 0 1 1 0 1 1 0 1 0 0 1 0 1 0 1 1 1 0 1 0 0 0 0 1 1 1 0
 0 0 1 1 0 0 1 0 1 0 0 0 1 1 1 1 1 0 0 0 1 1 0 1 1 1 0 0 1 1 0 1 0 0 1 0 1
 1 1 1 1 0 0 0 1 1 1 1 0 0 0 0 0 0 1 1 1 0 1 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0
 1 1 0 0 1 0 0 0 0 0 1 0 1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 0 0 0 1 1 0 1 1
 0 1 1 1 0 1 1 0 0 1 0 1 1 0 1 1 0 0 1 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 1
 1 1 1 1 0 0 1 0 1 0 1 1 1 0 0 0 0 1 0 1 1 1 1 0 0 1 1 0 1 1 1 0 0 1 1 0 1
 0 1 0 0 0 1 1 0 1 1 0 1 1 1 1 0 1 1 1 0 0 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 0
 0 0 1 0 0 1 1 0 0 0 0 1 1 0 0 1 1 0 0 0 1 1 1 0 0 0 1 1 0 1 1 1 1 1 0 1 1
 1 1 1 1 1 1 0 1 0 1 1 1 0 0 0 1 1 0 0 1 0 1 1 0 0 0 0 0 0 0 1 1 1 0 1 1 1
 1 0 0 1 1 0 0 1 0 0 1 1 0 1 0 1 1 0 1 0 0 0 0 1 0 1 0 0 1 0 0 1 1 1 1 0 0
 1 0 1 0 0 1 0 0 0 1 0 1 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 0 1 1 0 1 0 0 0 0 1
 0 0 1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0 0 0
 1 0 1 1 1 1 0 0 0 1 0 1 0 1 0 1 1 1 0 0 1 0 0 1 0 1 0 0 0 0 0 1 1 0 1 1
 0 1 1 0 1 1 0 1 1 0 0 0 0 0 1 0 0 0 1 0 0 1 0 0 0 1 1 1 0 1 1 1 0 0 0 0
 0 0 1 0 1 1 1 1 1 0 0 0 0 0 1 0 1 0 1 0 1 1 1 1 0 0 0 1 0 1 0 1 1 1 0 1 1
 0 0 1 0 1 0 0 1 0 1 1 0 0 1 1 1 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
 1 0 0 0 1 1 0 0 0 0 0 1 1 1 0 0 0 0 1 0 1 0 0 0 0 1 1 1 0 1 1 1 0 1 1 1 1
 0 1 1 0 0 1 1 0 0 1 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 1 0 1 1 1 0 0 1 0
 0 0 1 1 0 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 1 1 1 1 0 0 0 0 0 0 0 0 1 0 0 1
 0 0 1 1 1 1 0 1 0 0 1 0 0 1 0 0 1 0 0 1 0 1 1 1 0 1 0 1 0 1 0 1 0 1 1 1
 1 0 0 0 1 1 1 1 0 1 1 1 1 0 1 0 1 1 0 1 0 0 1 0 1 0 0 1 0 1 1 1 0 0 1 0 0
 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 0 1 0 1 0 1 1 0 0 0 1 0 0 1 1 1 1 0 1
 0 1 0 1 0 0 0 0 1 0 0 1 0 0 0 1 0 0 1 0 1 0 1 0 1 0 0 1 0 1 0 1 1 1 0 1 1
 0 0 1 1 0 1 0 1 1 0 0 1 0 1 1 1 1 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 0 0 0 1
 0 1 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0 1 0 0 1 1 0 1 0 1
 1 1 0 1 0 1 1 1 1 1 1 1 1 0 1 1 1 0 0 1 0 0 1 0 1 0 1 0 0 1 0 1 1 0 0 1 0 0
 0 0 1 0 1 1 0 0 0 1 0 0 0 0 1 1 1 0 0 0 1 1 0 0 1 0 1 0 0 1 0 0 0 1 0 1 0 0
 0]
```

C:\Users\HP\Anaconda3\lib\site-packages\sklearn\preprocessing\label.py:235: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)

```
In [19]: print(df['race/ethnicity'].value_counts())
```

```
group C    319
group D    262
group B    190
group E    140
group A     89
Name: race/ethnicity, dtype: int64
```

```
In [20]: df_Lunch = pd.get_dummies(df['lunch'])
df_new = pd.concat([df, df_Lunch], axis=1)
print(df_new)
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	
..	...	...	...	...	
995	female	group E	master's degree	standard	
996	male	group C	high school	free/reduced	
997	female	group C	high school	free/reduced	
998	female	group D	some college	standard	
999	female	group D	some college	free/reduced	

	test preparation course	math score	reading score	writing score	\
0	none	72	72	NaN	
1	completed	69	90	88.0	
2	none	90	95	93.0	
3	none	47	57	44.0	
4	none	76	78	75.0	
..	...	...	...	...	
995	completed	88	99	95.0	
996	none	62	55	55.0	
997	completed	59	71	65.0	
998	completed	68	78	77.0	
999	none	77	86	86.0	

	free/reduced	standard
0	0	1
1	0	1
2	0	1
3	1	0
4	0	1
..	...	...
995	0	1
996	1	0
997	1	0
998	0	1
999	1	0

```
[1000 rows x 10 columns]
```

```
In [ ]:
```